

● EASY

● ADVANCED MATH

● ANSWER KEY

# Nonlinear equations in one variable and systems of equations in two variables

01

10 answers · Rationale-rich · Teach from doc

**Q1****A****A.** 432

Choice A is correct. The first equation in the given system of equations is  $x = 3$ . Substituting 3 for  $x$  in the second equation in the given system of equations yields  $y = 15 - 3^2$ , or  $y = 144$ . Substituting 3 for  $x$  and 144 for  $y$  in the expression  $xy$  yields  $(3)(144)$ , or 432. Therefore, the value of  $xy$  is 432.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q2****9**

The correct answer is 9. Dividing both sides of the given equation by 5 yields  $|x| = 9$ . By the definition of absolute value, if  $|x| = 9$ , then  $x = 9$  or  $x = -9$ . Therefore, the two solutions to the equation  $5x = 45$  are 9 and -9. It follows that the positive solution to the given equation is 9.

**Q3****A****A.** 16

Choice A is correct. It's given that the graphs of the given equations intersect at the point  $x, y$  in the  $xy$ -plane. It follows that  $x, y$  represents a solution to the system consisting of the given equations. The first equation given is  $x = 49$ . Substituting 49 for  $x$  in the second equation given,  $y = \sqrt{x} + 9$ , yields  $y = \sqrt{49} + 9$ , which is equivalent to  $y = 7 + 9$ , or  $y = 16$ . It follows that the graphs of the given equations intersect at the point 49, 16. Therefore, the value of  $y$  is 16.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q4****B****B.** Exactly two

Choice B is correct. The number of solutions to a quadratic equation of the form  $ax^2 + bx + c = 0$ , where  $a$ ,  $b$ , and  $c$  are constants, can be determined by the value of the discriminant,  $b^2 - 4ac$ . If the value of the discriminant is positive, then the quadratic equation has exactly two distinct real solutions. If the value of the discriminant is equal to zero, then the quadratic equation has exactly one real solution. If the value of the discriminant is negative, then the quadratic equation has zero real solutions. In the given equation,  $3x^2 - 15x + 18 = 0$ ,  $a = 3$ ,  $b = -15$ , and  $c = 18$ . Substituting 3 for  $a$ , -15 for  $b$ , and 18 for  $c$  in  $b^2 - 4ac$  yields  $(-15)^2 - 4(3)(18)$ , or 9. Since the value of the discriminant is positive, the given equation has exactly two distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q5****B**

B.  $2x - 3y > 4$

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Choice B is correct. Both sides of the given inequality can be divided by 3 to yield  $2x - 3y > 4$ .

Choices A, C, and D are incorrect because they are not equivalent to (do not have the same solution set as) the given inequality. For example, the ordered pair  $(0, -1.5)$  is a solution to the given inequality, but it is not a solution to any of the inequalities in choices A, C, or D.

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**Q6****B**

B. 4

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Choice B is correct. Subtracting 61 from each side of the given equation yields  $p = 4$ . By the definition of absolute value, if  $p = 4$ , then  $p = 4$  or  $p = -4$ . Of the given choices, 4 is a solution to the given equation.

Choice A is incorrect. This is the quotient, not the difference, of 65 and 61.

Choice C is incorrect. This is the sum, not the difference, of 65 and 61.

Choice D is incorrect and may result from conceptual or calculation errors.

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**Q7****B****B.** 2

Choice B is correct. Subtracting  $x^2$  from both sides of the given equation yields  $x^2 - 4 = 0$ . Adding 4 to both sides of the equation gives  $x^2 = 4$ . Taking the square root of both sides of the equation gives  $x = 2$  or  $x = -2$ . Therefore,  $x = 2$  is one solution to the original equation.

Alternative approach: Subtracting  $x^2$  from both sides of the given equation yields  $x^2 - 4 = 0$ . Factoring this equation gives  $x^2 - 4 = (x + 2)(x - 2) = 0$ , such that  $x = 2$  or  $x = -2$ . Therefore,  $x = 2$  is one solution to the original equation.

Choices A, C, and D are incorrect and may be the result of computation errors.

**Q8****C****C.** 4, 5

Choice C is correct. The solution to the system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear equation and the nonlinear equation shown intersect at the point 4, 5. Thus, the solution to the system is 4, 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q9****D****D.  $-3$** 

Choice D is correct. If  $(x+5)^2 = 4$ , then taking the square root of each side of the equation gives  $x+5 = 2$  or  $x+5 = -2$ . Solving these equations for  $x$  gives  $x = -3$  or  $x = -7$ . Of these,  $-3$  is the only solution given as a choice.

Choice A is incorrect and may result from solving the equation  $x+5 = 4$  and making a sign error. Choice B is incorrect and may result from solving the equation  $x+5 = 4$ . Choice C is incorrect and may result from finding a possible value of  $x+5$ .

**Q10****C****C.  $-\frac{7}{2}, \frac{9}{2}$** 

Choice C is correct. The solution to a system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at a point with an  $x$ -coordinate between  $-4$  and  $-3$  and a  $y$ -coordinate between  $4$  and  $5$ . Of the given choices, only  $-\frac{7}{2}, \frac{9}{2}$  has an  $x$ -coordinate between  $-4$  and  $-3$  and a  $y$ -coordinate between  $4$  and  $5$ .

Choice A is incorrect. This is the  $y$ -intercept of the graph of the linear function.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.

**Notes**

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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